

Assignment 7

Q1

```
#!/bin/bash
echo "Login Name   : $LOGNAME"
echo "Home Directory : $HOME"
```

Q2

```
#!/bin/bash
```

```
while true
do
    echo "1. Date"
    echo "2. Cal"
    echo "3. Ls"
    echo "4. Pwd"
    echo "5. Exit"
    echo -n "Enter your choice: "
    read ch

    case $ch in
        1) date ;;
        2) cal ;;
        3) ls ;;
        4) pwd ;;
        5) exit 0 ;;
        *) echo "Invalid choice" ;;
    esac

    echo
done
```

Q3

```
#!/bin/bash
```

```
echo -n "Enter a name: "
read name

if [ -f "$name" ]
then
    echo "$name is a file. Size = $(wc -c < "$name") bytes"
elif [ -d "$name" ]
then
    echo "$name is a directory. Contents:"
    ls "$name"
else
    echo "$name does not exist"
fi
```

Q4

```
#!/bin/bash
```

```
echo -n "Enter a number: "
read n
```

```
factors=$(factor "$n")

if [ "$factors" = "$n: $n" ]
then
    echo "$n is a prime number"
else
    echo "$n is not a prime number"
fi
```

Q5

```
#!/bin/bash
```

```
echo -n "Enter three numbers: "
read a b c
```

```
if [ "$a" -ge "$b" ] && [ "$a" -ge "$c" ]
then
    echo "$a is the greatest"
elif [ "$b" -ge "$a" ] && [ "$b" -ge "$c" ]
then
    echo "$b is the greatest"
else
    echo "$c is the greatest"
fi
```

Q6

```
#!/bin/bash
```

```
echo -n "Enter a year: "
read y
```

```
if [ $((y % 400)) -eq 0 ] || ( [ $((y % 4)) -eq 0 ] && [ $((y % 100)) -ne 0 ] )
then
    echo "$y is a leap year"
else
    echo "$y is not a leap year"
fi
```

Q7

```
#!/bin/bash
```

```
echo -n "Enter a number: "
read n
```

```
if [ "$n" -gt 0 ]
then
    echo "$n is positive"
elif [ "$n" -lt 0 ]
then
    echo "$n is negative"
else
    echo "The number is zero"
fi
```

Q8

```
#!/bin/bash
```

```
echo -n "Enter a number: "  
read n
```

```
for i in 1 2 3 4 5 6 7 8 9 10  
do  
    echo "$n x $i = $((n * i))"  
done
```

Q9

```
#!/bin/bash
```

```
echo -n "Enter a number: "  
read n
```

```
f=1  
i=1
```

```
while [ "$i" -le "$n" ]  
do  
    f=$((f * i))  
    i=$((i + 1))  
done
```

```
echo "Factorial of $n = $f"
```

Q10

```
#!/bin/bash
```

```
echo -n "Enter number of terms: "  
read n
```

```
a=0  
b=1
```

```
echo "Fibonacci series:"
```

```
for (( i=0; i<n; i++ ))  
do  
    echo -n "$a "  
    c=$((a + b))  
    a=$b  
    b=$c  
done
```

```
echo
```

Q11

```
#!/bin/bash
```

```
echo -n "Enter basic salary: "  
read basic
```

```
da=$((echo "$basic * 40 / 100" | bc -l))  
hra=$((echo "$basic * 20 / 100" | bc -l))
```

```
gross=$(echo "$basic + $da + $hra" | bc -l)
```

```
printf "DA   = %.2f\n" "$da"
printf "HRA  = %.2f\n" "$hra"
printf "Gross = %.2f\n" "$gross"
```

Q12

```
#!/bin/bash
```

```
if [ $# -ne 1 ]
then
    echo "Usage: $0 filename"
    exit 1
fi
```

```
if [ -e "$1" ]
then
    echo "Last modification time of $1:"
    stat -c %y "$1"
else
    echo "File '$1' does not exist"
fi
```

Q13

```
#!/bin/bash
```

```
echo "Hidden files in $(pwd):"
```

```
for f in .[*]
do
    if [ -f "$f" ]
    then
        echo "$f"
    fi
done
```

Q14

```
#!/bin/bash
```

```
echo "Executable files in $(pwd):"
```

```
for f in *
do
    if [ -f "$f" ] && [ -x "$f" ]
    then
        echo "$f"
    fi
done
```

Q15

```
#!/bin/bash
```

```
echo -n "Enter first file name: "
read f1
```

```
echo -n "Enter second file name: "
read f2

if [ ! -f "$f1" ]
then
    echo "$f1 does not exist"
    exit 1
fi

tr 'a-zA-Z' 'A-Za-z' < "$f1" >> "$f2"

echo "Contents of $f1 appended to $f2 in reverse case"
```

Q16

```
#!/bin/bash
```

```
echo "Welcome, $LOGNAME!"
echo "Contents of your home directory:"
ls "$HOME"
```

```
chmod +x ~/welcome.sh
```

```
echo 'bash ~/welcome.sh' >> ~/.bashrc
```

Q17

```
#!/bin/bash
```

```
for (( i=1; i<=5; i++ ))
do
    for (( j=1; j<=i; j++ ))
    do
        echo -n "*"
    done
    echo
done
```